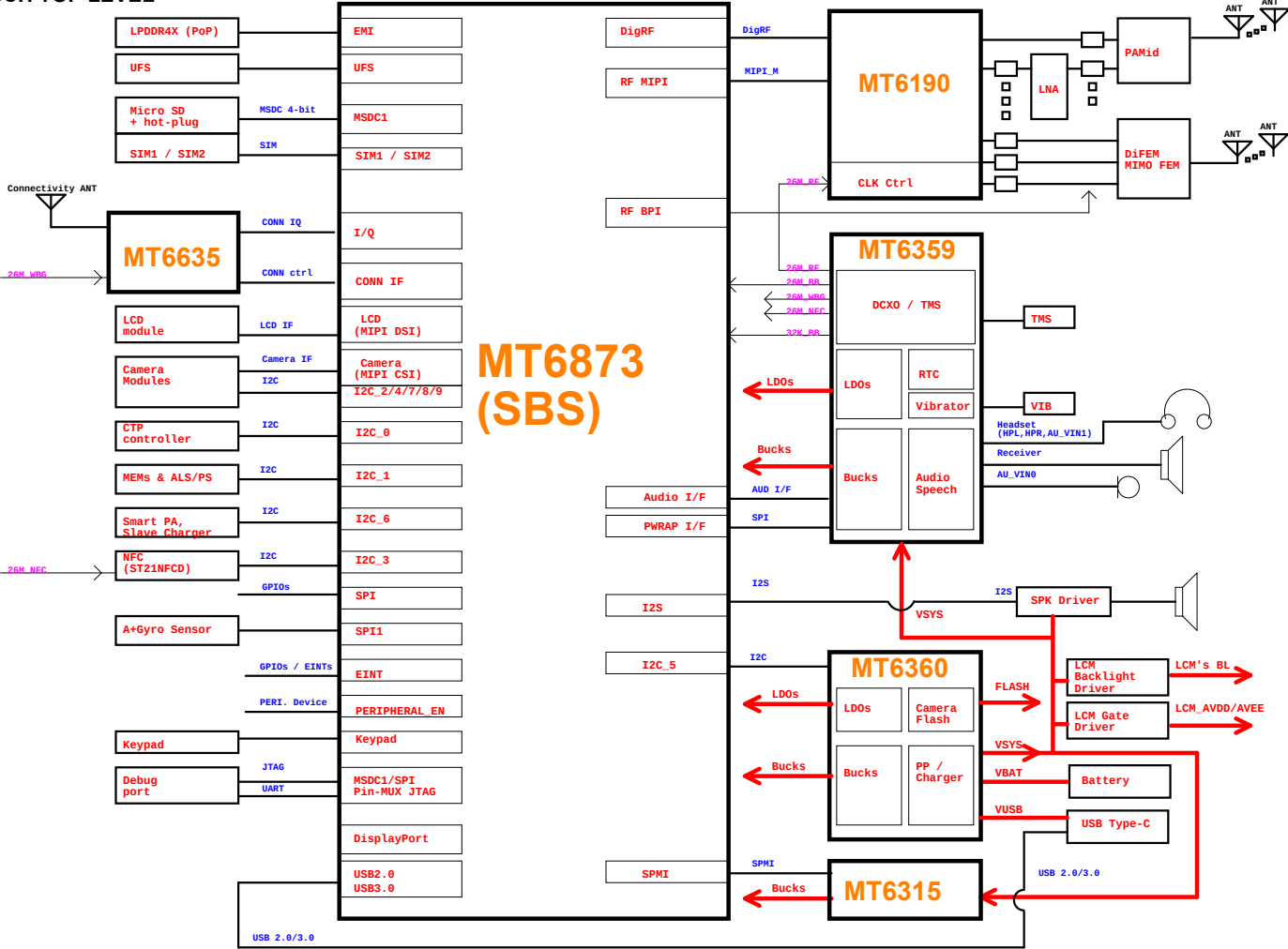
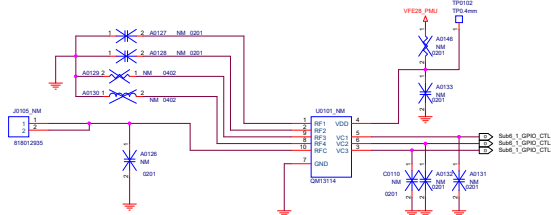
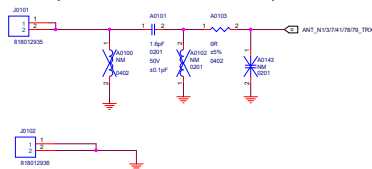


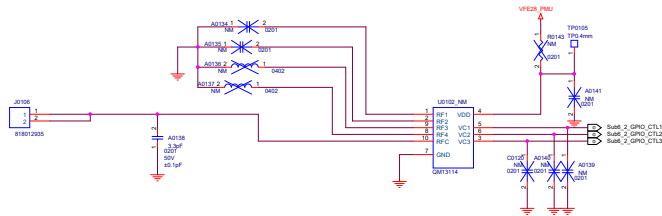
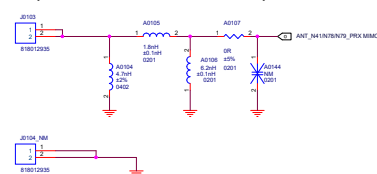
Project : MT6873, LPDDR4X (SBS)  
REF\_SCH TOP LEVEL



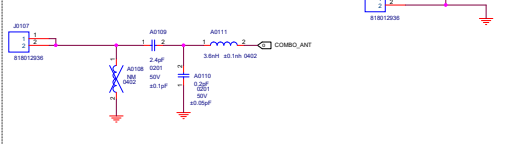
# ANT3(N1/3/7/41/78/79\_TRX)



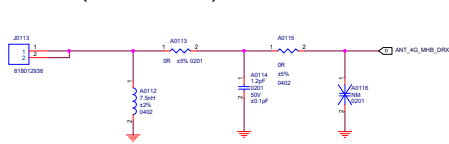
# ANT4(N41/N78/N79\_PRX MIMO)



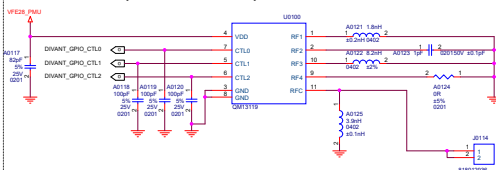
# Combo ANT



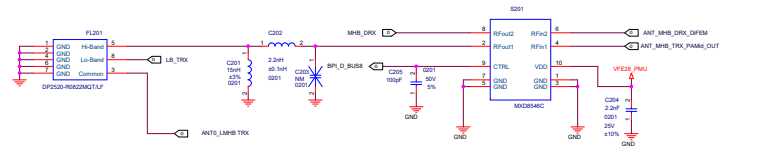
# ANT1(4G\_LMHB DRX)\_Feed



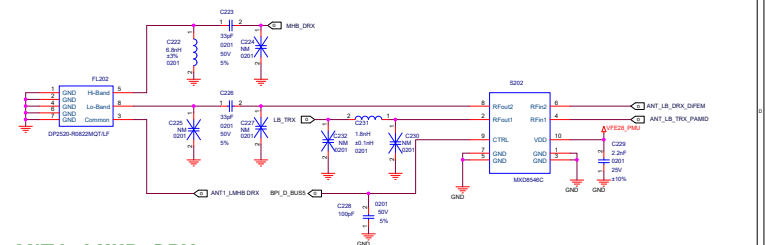
# ANT1(4G\_LMHB DRX)\_SW



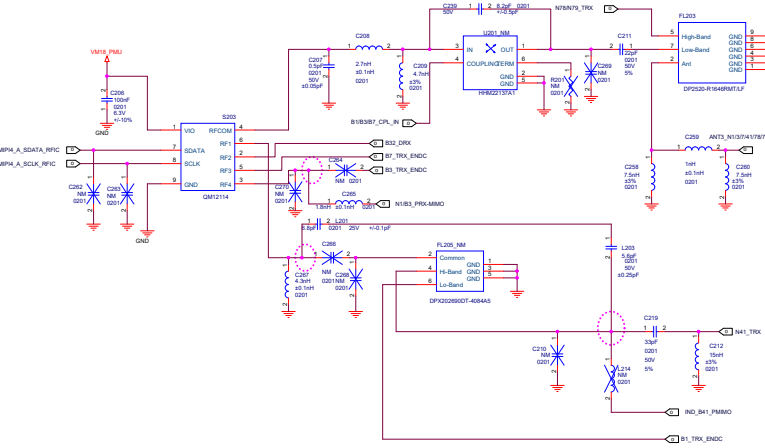
### ANT0\_LMHB TRX



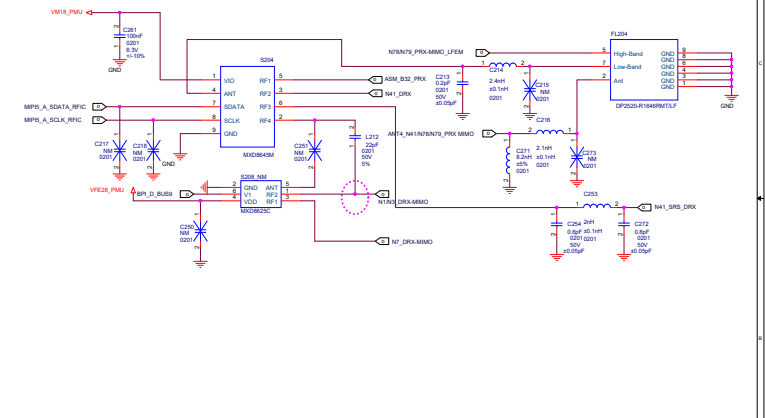
### ANT1\_LMHB DRX



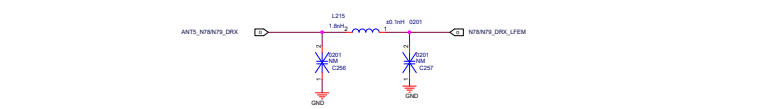
### ANT3\_N1/3/7/41/78/79 TRX



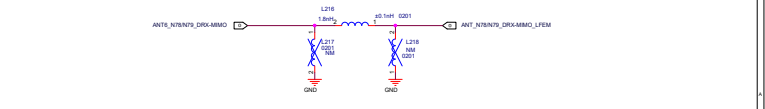
### ANT4\_N1/3 DRX4+N41/78/79 PRX2



### ANT5\_78/79 DRX3

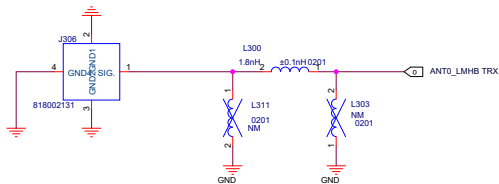


### ANT6\_78/79 DRX4

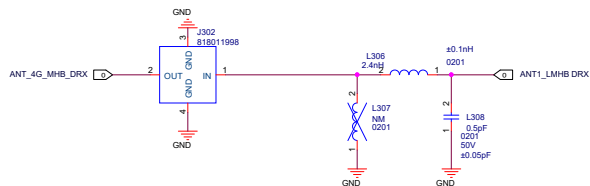


| IN PHONE                      |                                       |       |           |
|-------------------------------|---------------------------------------|-------|-----------|
| File<br>MT6653                |                                       |       |           |
| Size<br>A2                    | Document Number<br>Reserved(for ANT4) |       | Rev<br>P0 |
| Date: Monday, August 03, 2020 |                                       | Sheet | 2 of 73   |

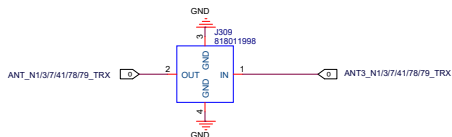
## ANT0(4G\_LMHB TRX)



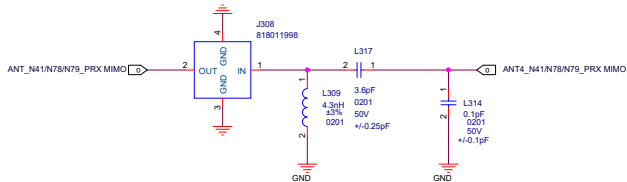
## ANT1(4G\_LMHB DRX)



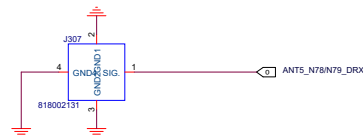
## ANT3(N1/3/7/41/78/79\_TRX)



## ANT4(N41/N78/N79\_PRX MIMO)



## ANT5(N78/N79\_DRX)

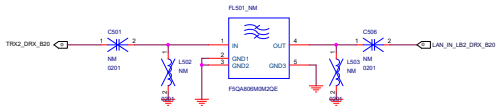


## ANT6(N78/N79\_DRX-MIMO)

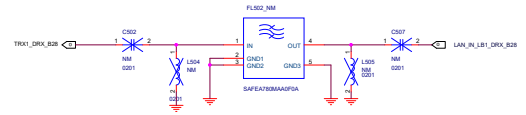


| MI PHONE |                         |                 |         |
|----------|-------------------------|-----------------|---------|
| Title    | MT0853                  |                 |         |
| Size     | AS                      | Document Number | Rev     |
|          |                         | RP_CONN         | PO      |
| Date:    | Monday, August 05, 2020 | Sheet           | 3 of 73 |

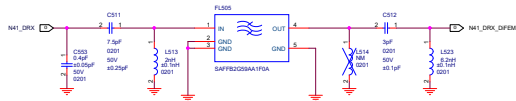




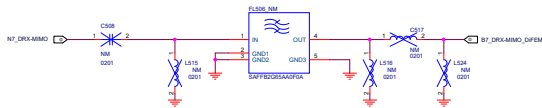
**B20 DRX**



**B28 DRX**

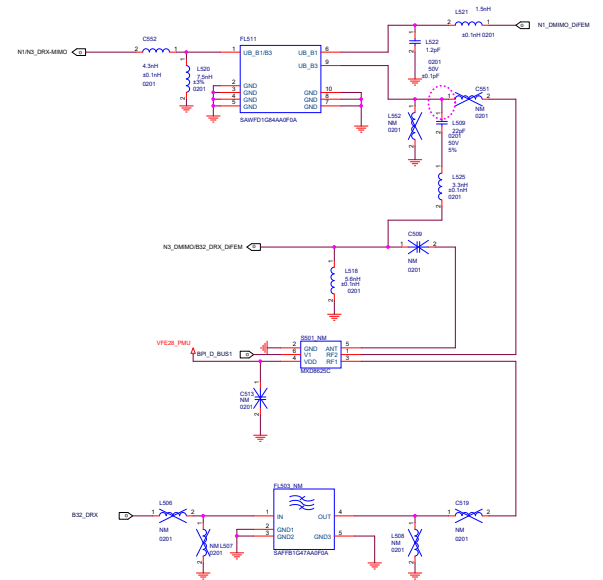


**N41 DRX**



**N7 DRX MIMO**

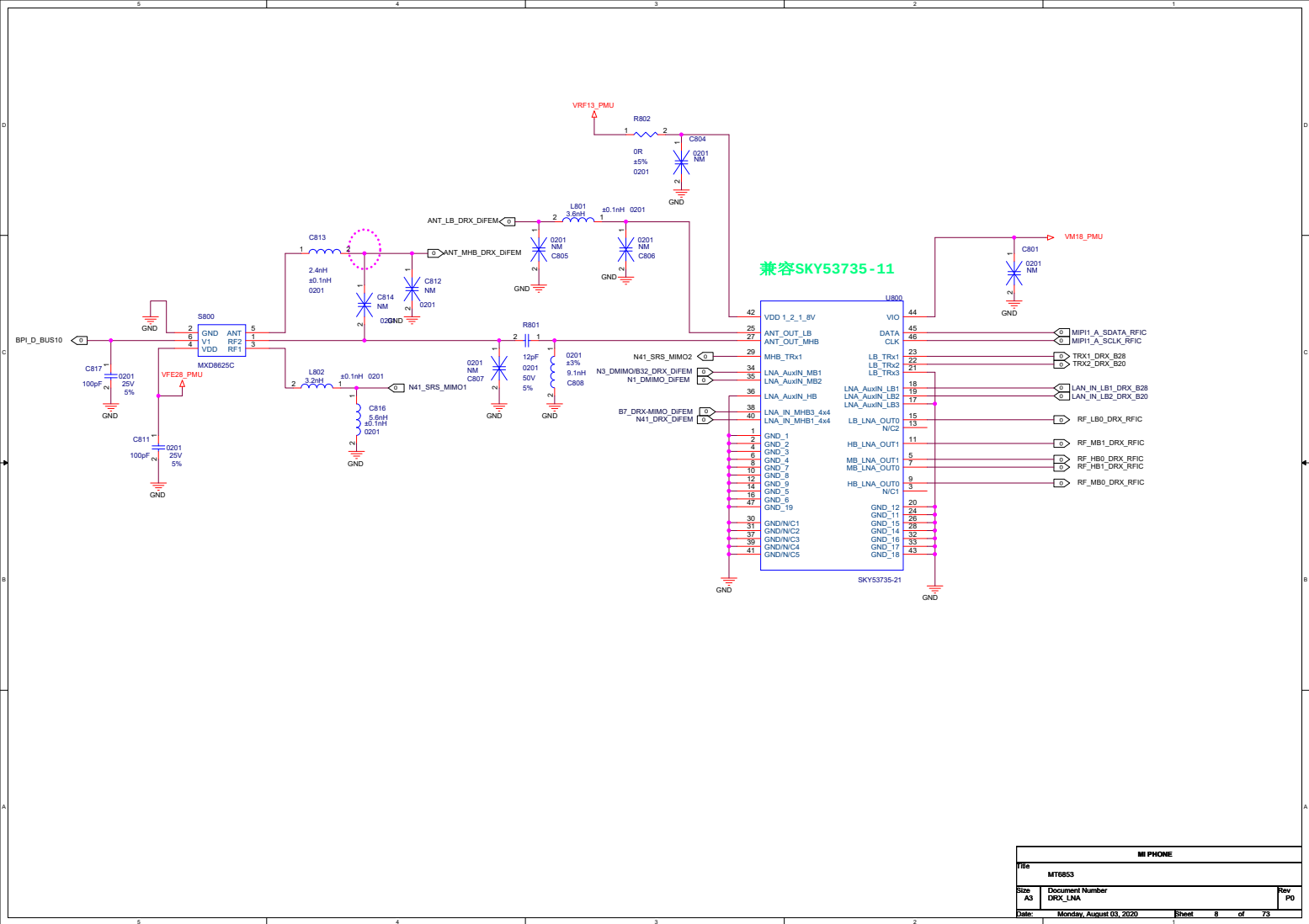
**N1/N3 DRX MIMO**



**B32 DRX**

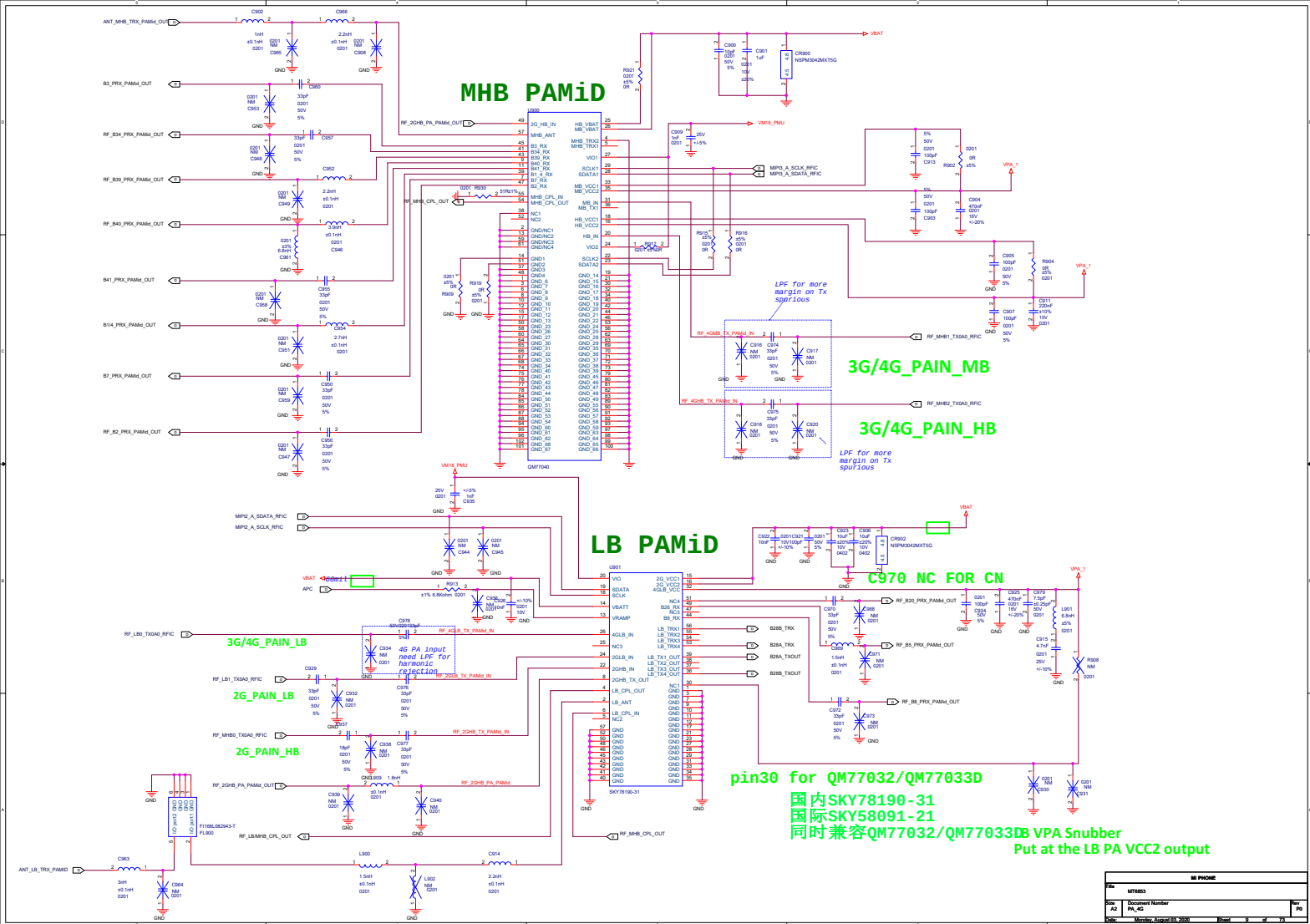
| M PHONE         |         |     |     |
|-----------------|---------|-----|-----|
| Rev             | M1855   |     |     |
| Document Number |         | Doc | IP  |
| A2              | Version | 1.0 | 1.0 |

|                               |                         |       |        |
|-------------------------------|-------------------------|-------|--------|
| MI PHONE                      |                         |       |        |
| Title MT6853                  |                         |       |        |
| Size A3                       | Document Number PRX_LNA |       | Rev P0 |
| Date: Monday, August 03, 2020 | Sheet 7                 | of 73 |        |



| MI PHONE |                         |                 |       |
|----------|-------------------------|-----------------|-------|
| Title    | MT8853                  |                 |       |
| Size     | AS                      | Document Number | Rev   |
|          |                         | DRX_LNA         | PD    |
| Date:    | Monday, August 05, 2020 |                 |       |
|          | Sheet                   | 8               | of 73 |

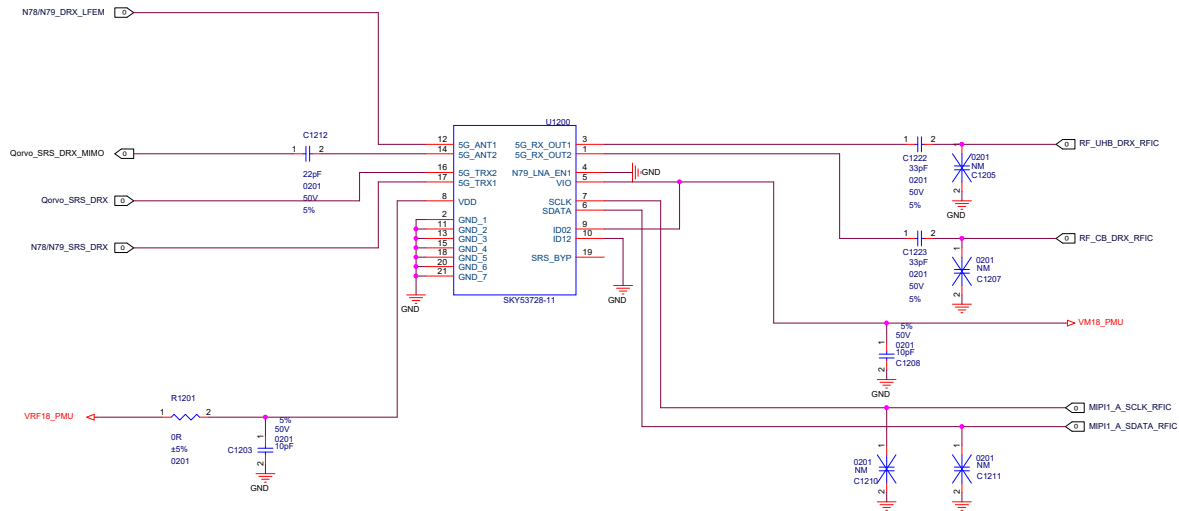




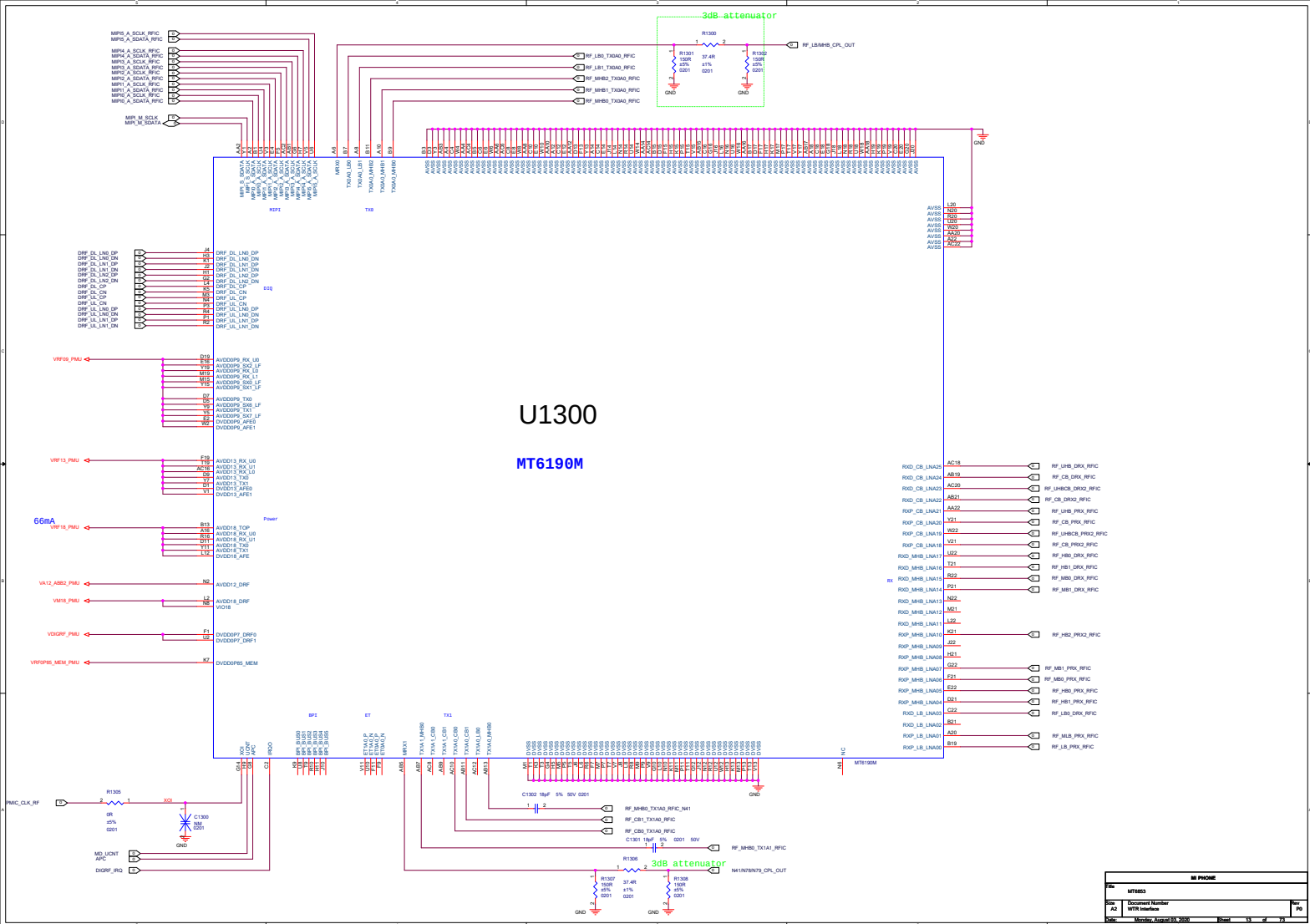
| IN PACKAGE                             |                 |  |       |
|----------------------------------------|-----------------|--|-------|
| Rev                                    | MT9833          |  |       |
| Doc                                    | Document Number |  | Rev   |
| PL_A2                                  |                 |  | PL_A2 |
| Date: Monday, August 03, 2009 10:00 AM |                 |  |       |

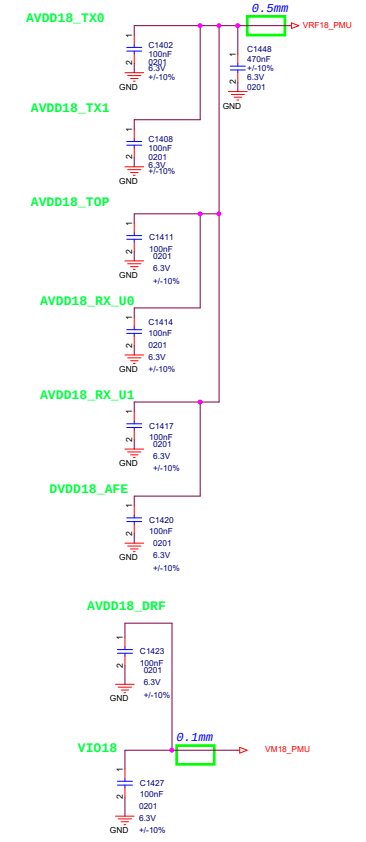
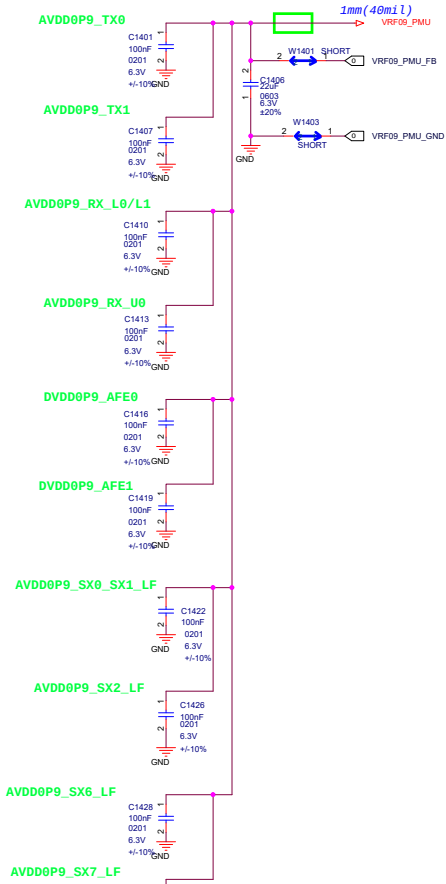
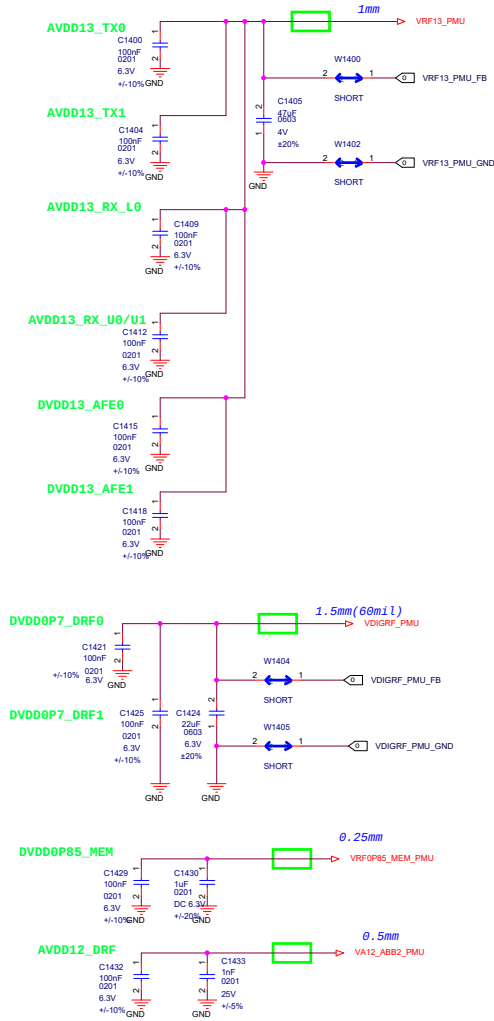




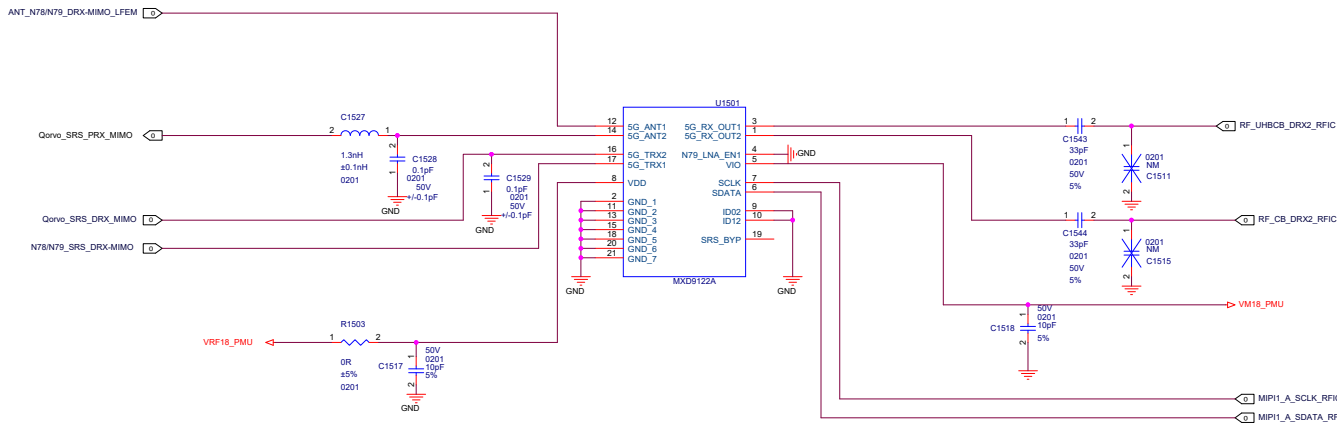
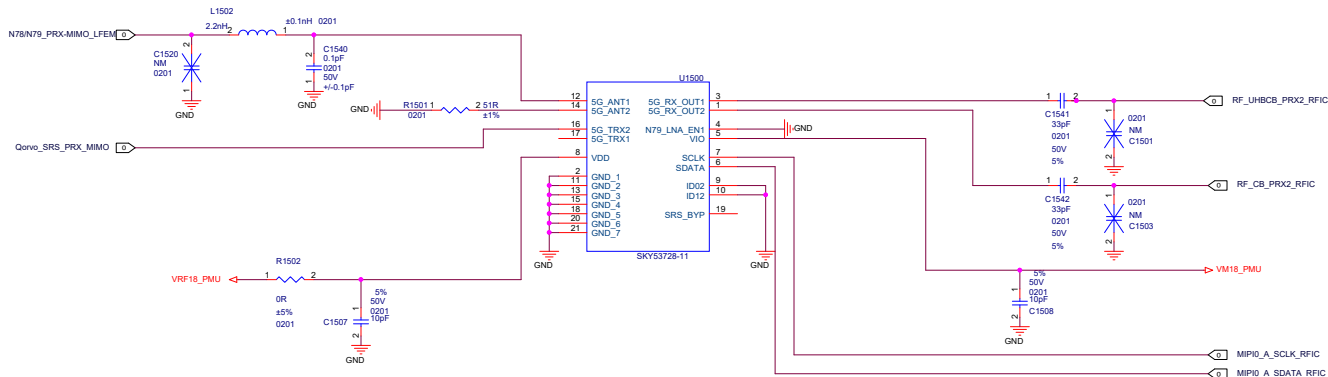


| MI PHONE |                         |                |
|----------|-------------------------|----------------|
| Title    | MT6853                  |                |
| Size     | Document Number         | Rev            |
| A3       | PA_5G_TX1_N78_N79       |                |
| Date:    | Monday, August 03, 2020 | Sheet 12 of 73 |





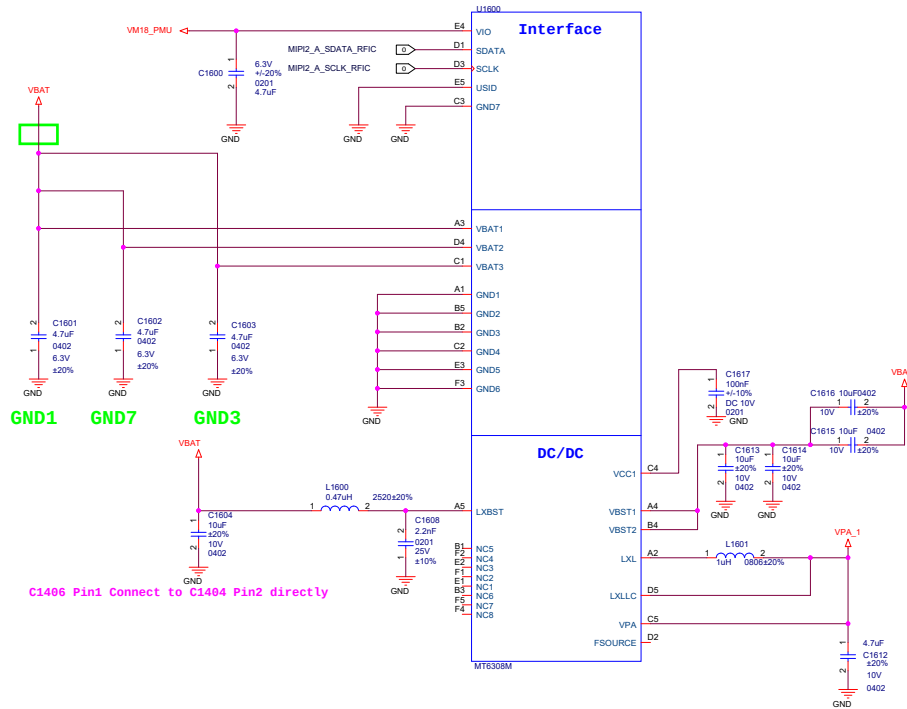
| MT8853 |                         |                 |         |
|--------|-------------------------|-----------------|---------|
| File   | MT8853                  |                 |         |
| Size   | AS                      | Document Number | WTR PMW |
| Date:  | Monday, August 05, 2020 |                 |         |
| Sheet  | 14                      | of              | 73      |



| MI PHONE |                         |                 |             |
|----------|-------------------------|-----------------|-------------|
| Title    | MT0853                  |                 |             |
| Size     | AS                      | Document Number | SG_MIMO_LNA |
| Date:    | Monday, August 03, 2020 | Sheet           | 15 of 73    |

# NR\_PAMid\_n41\_SatPA\_ET1\_NR1/ET Ref design

## L+MH\_PAMiId\_ET

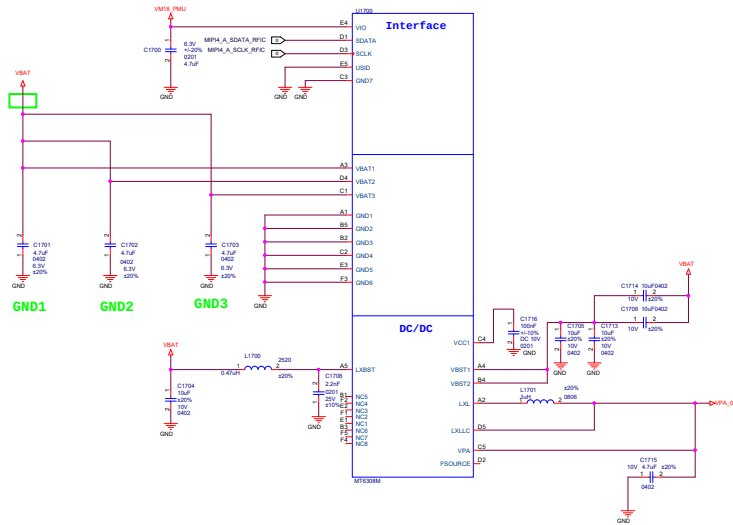


GRM155R61A105KE15 (0402): 0.73uF @ 4.2V

| MI PHONE |                         |       |          |
|----------|-------------------------|-------|----------|
| Title    | MT6853                  |       |          |
| Size     | Document Number         | Rev   |          |
| A3       | RP_402_D00C             | PD    |          |
| Date:    | Monday, August 03, 2020 | Sheet | 16 of 73 |

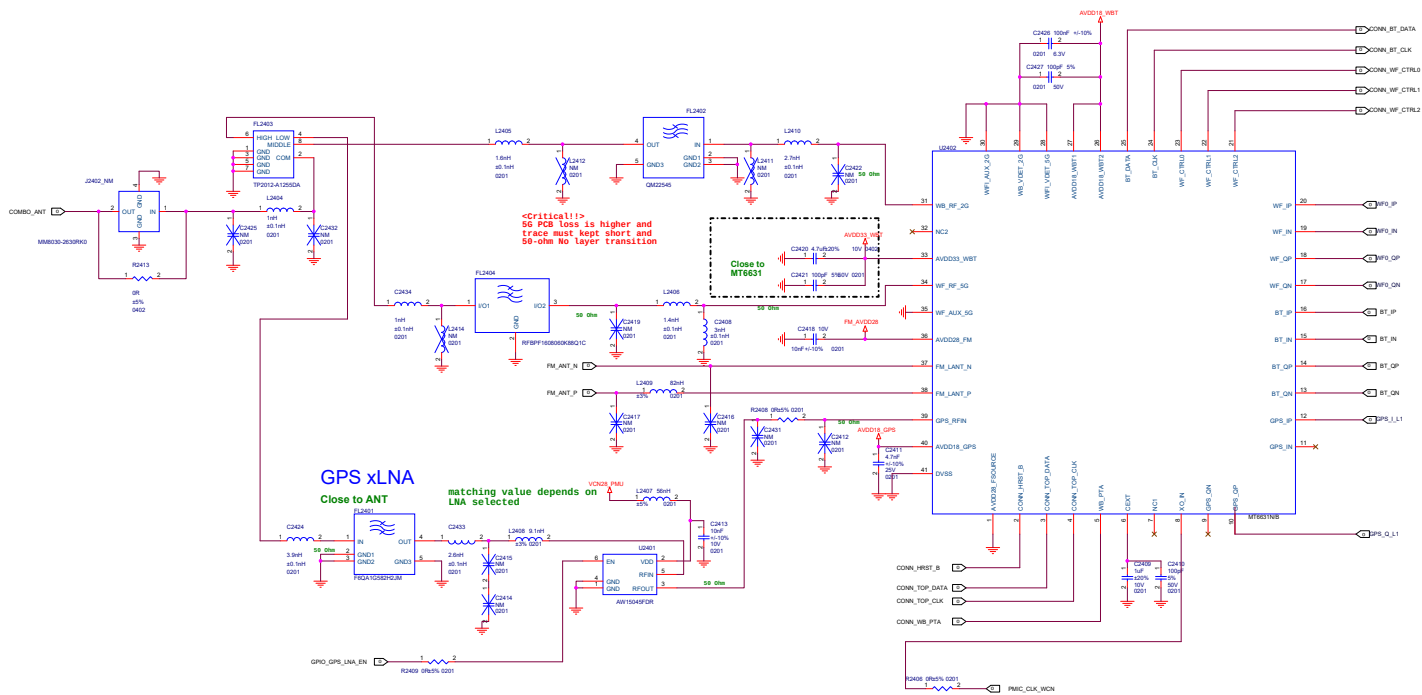


# NR\_PAMid + n41 PAD /APT Ref Design



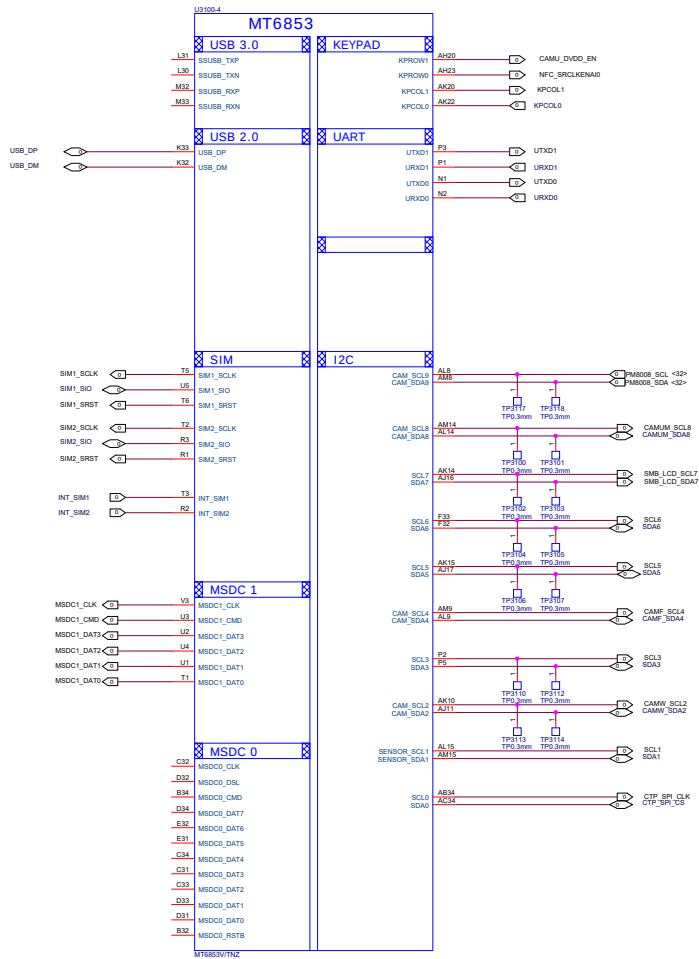
| IN PROCESS |        |     |    |
|------------|--------|-----|----|
| Rev        | 000000 |     |    |
| Rev        | 000000 | Rev | 00 |
| Rev        | 000000 | Rev | 00 |





Note: For MT6631 need to configure the VCN33\_2\_PMU to 2.8V by PMIC SW.

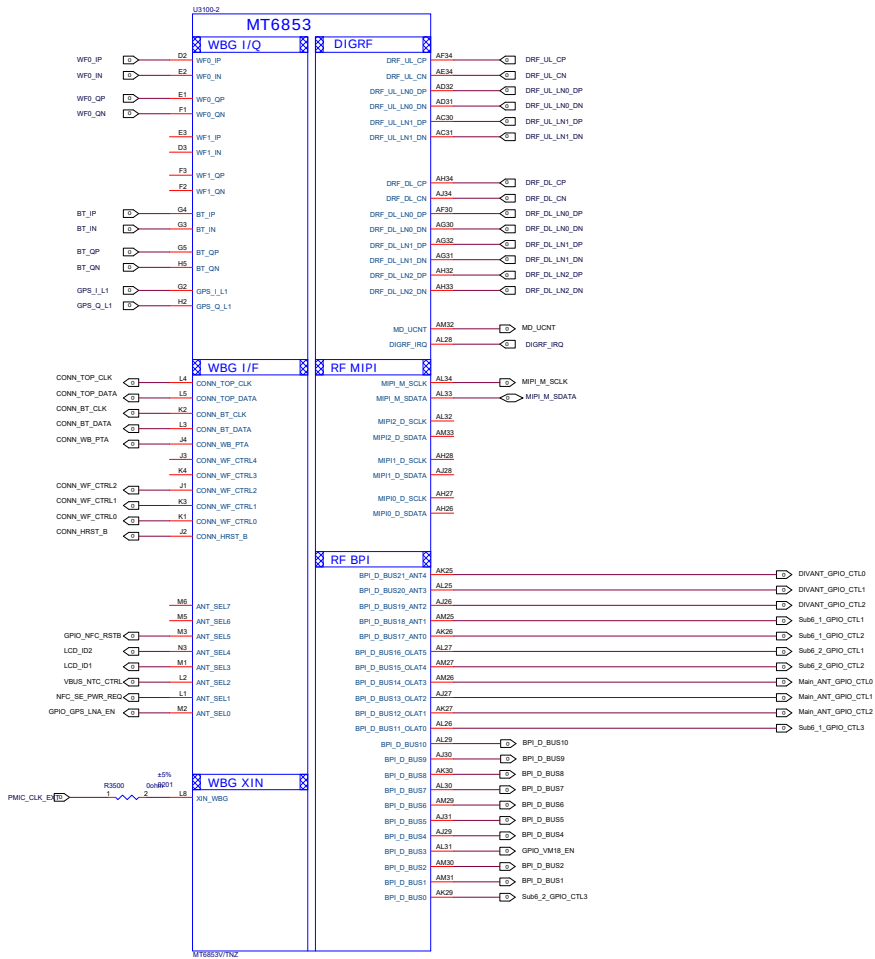
|     |                 |     |
|-----|-----------------|-----|
| Rev | MT6631          | Rev |
| Rev | Document Number | Rev |
| Rev | Rev             | Rev |
| Rev | Rev             | Rev |



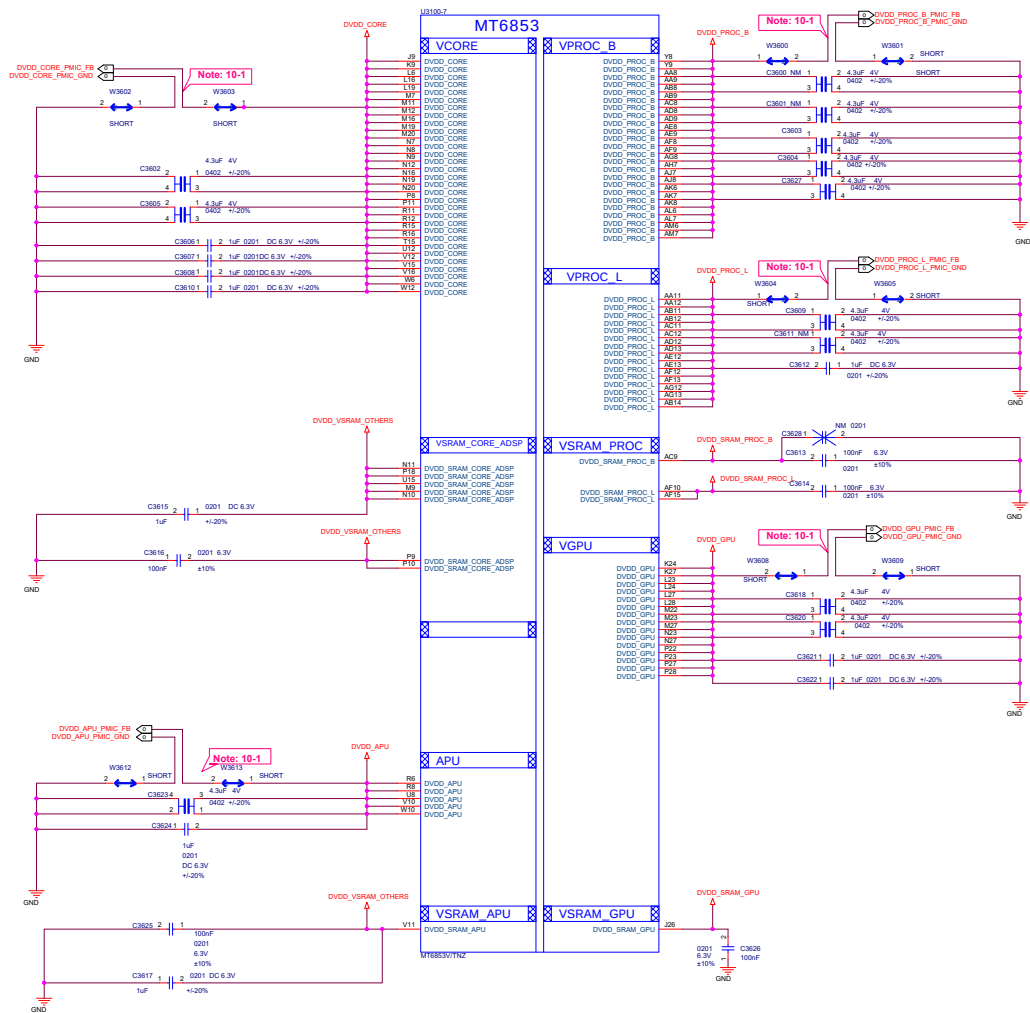










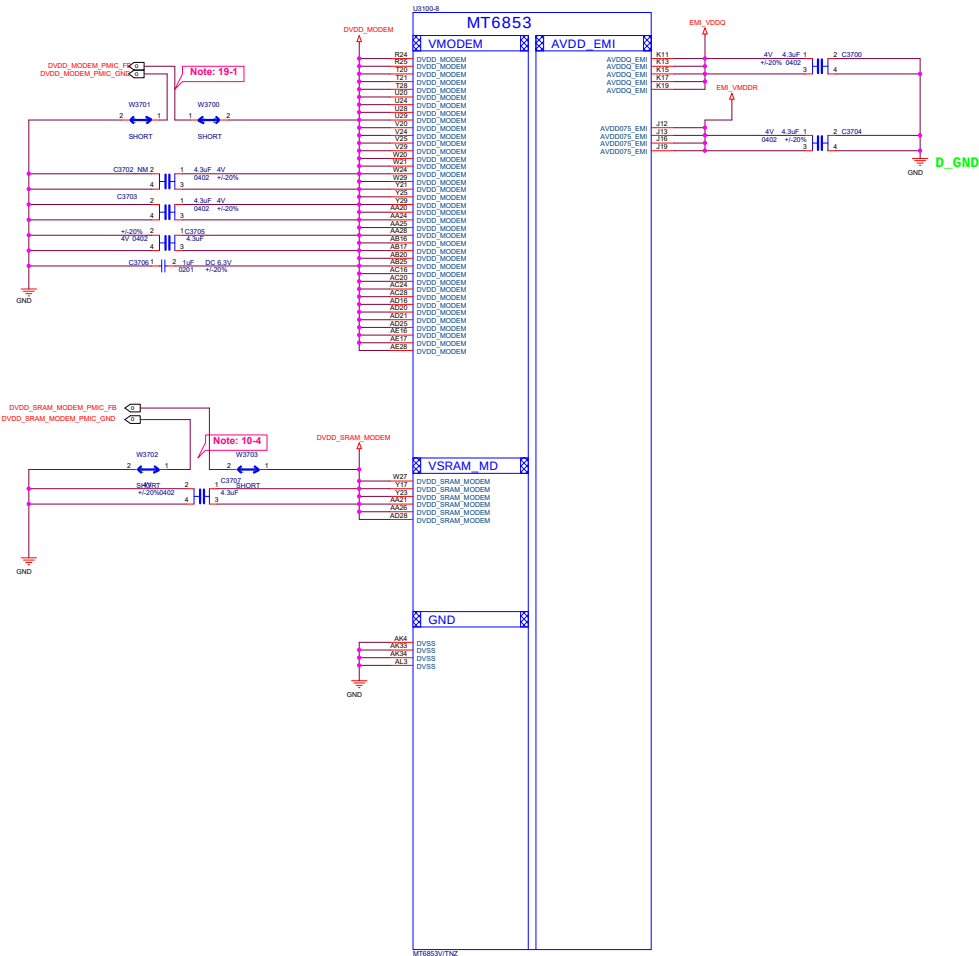


## Schematic design notice of "10\_BB\_POWER\_PDN" page.

Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.

Remote sense trace with GND shielding to PMIC (Differential)

| MT6853 |                         |       |          |
|--------|-------------------------|-------|----------|
| File   | MT6853                  |       |          |
| Size   | Document Number         |       |          |
| C      | MT6853_PWR1             |       |          |
| Valid  | Monday, August 28, 2020 | Sheet | 35 of 73 |



# Schematic design notice of "11\_BB\_POWER\_PDN" page.

Note 11-1: Differential pair of PMIC remote sense must be close to BB's ball.  
Remote sense trace with GND shielding to PMIC (Differential)

| MT6853 |                         |       |                 |
|--------|-------------------------|-------|-----------------|
| File   | MT6853                  | Size  | Document Number |
| Size   | MT6853_PWR2             | Rev   |                 |
| Date   | Monday, August 01, 2023 | Sheet | 37 of 73        |

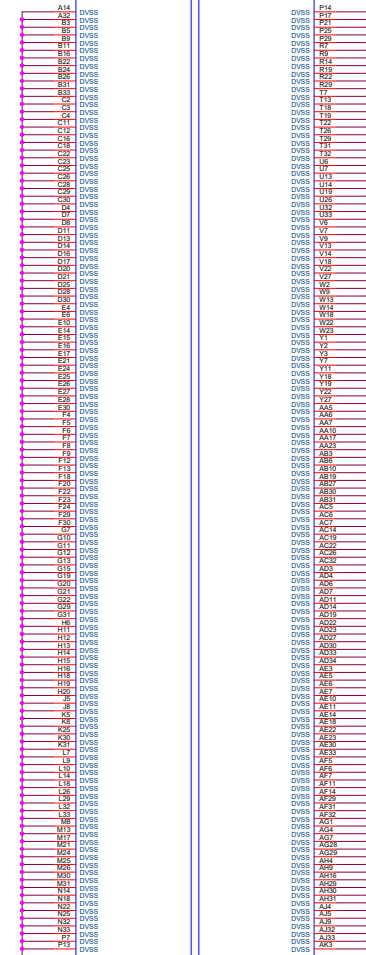


U3100-9

MT6853

GND

GND



MT6853/71N2

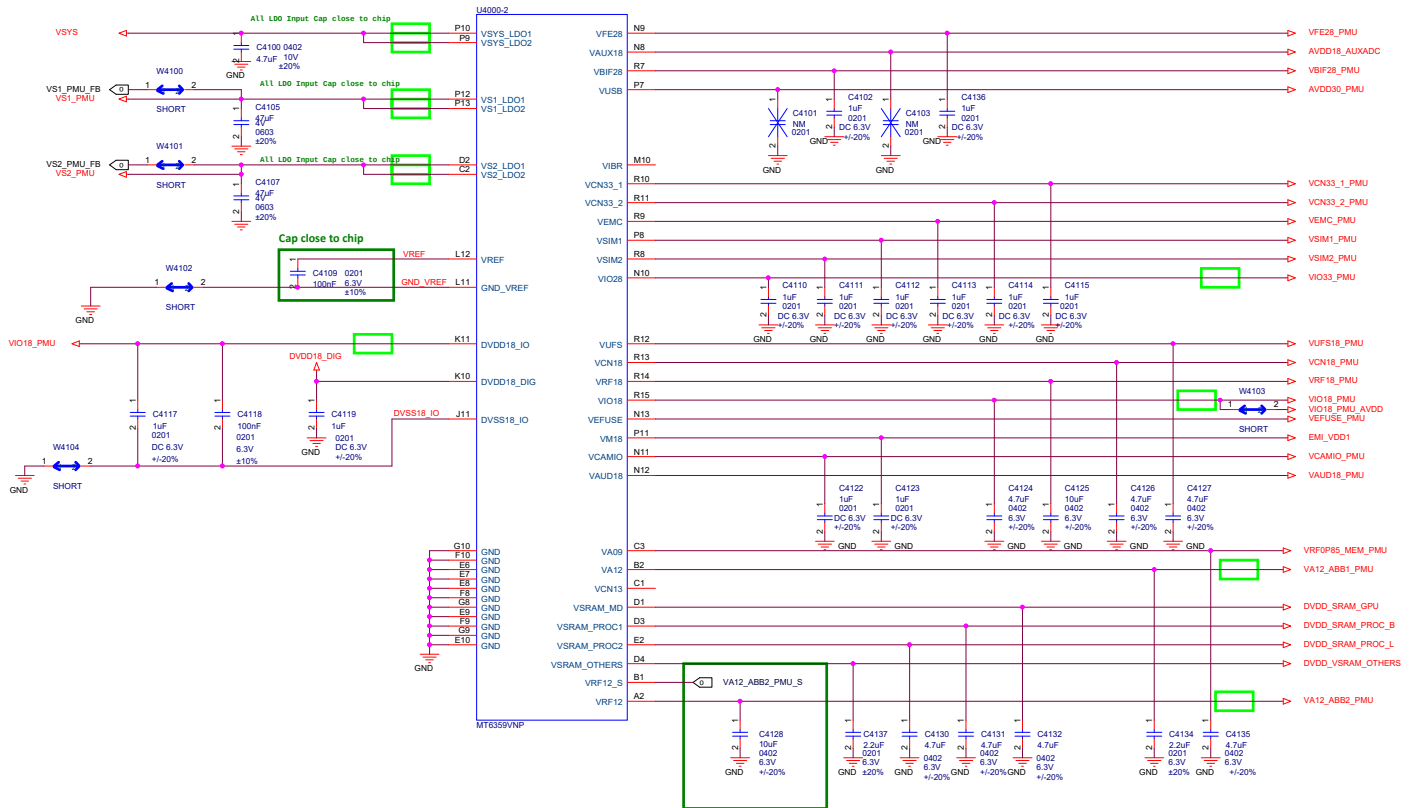


GND

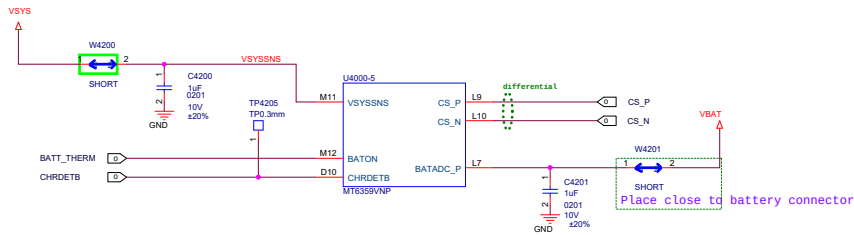
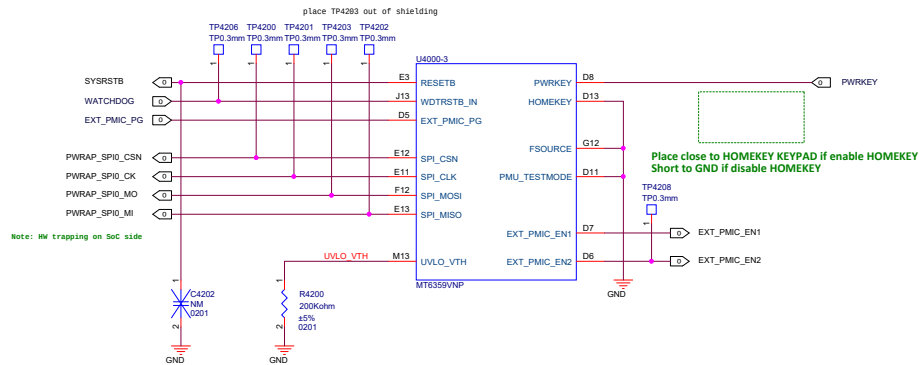
| MT6853 |                         |       |          |
|--------|-------------------------|-------|----------|
| File   | MT6853                  |       |          |
| Size   | Document Number         | Rev   |          |
| C      | MT6853_GND              |       |          |
| Date   | Monday, August 01, 2005 | Sheet | 39 of 73 |



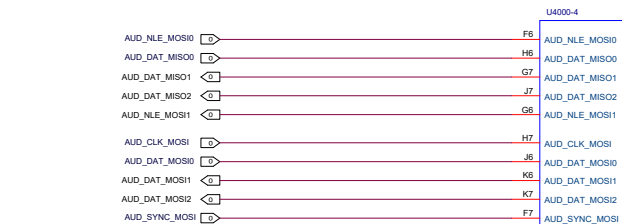
1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2mohm)  
=> value and placement of Cap, please refer design notice



| MI PHONE |                         |       |          |
|----------|-------------------------|-------|----------|
| Title    | MT6853                  |       |          |
| Size     | Document Number         | Rev   |          |
| A3       | MT6359_LDO              |       |          |
| Date:    | Monday, August 03, 2020 | Sheet | 41 of 73 |



| MI PHONE |                         |       |          |
|----------|-------------------------|-------|----------|
| Title    | MT6853                  |       |          |
| Size     | MT6359VNP               | Rev   |          |
| Date     | Monday, August 03, 2020 | Sheet | 42 of 73 |



For P-N pair: differential pair & GND shielding!

AU\_VIN0\_P L1 AU\_VIN0\_P

For P-N pair: differential pair & GND shielding!

AU\_VIN0\_N L2 AU\_VIN0\_N

For P-N pair: differential pair & GND shielding!

AU\_VIN1\_P M2 AU\_VIN1\_P

For P-N pair: differential pair & GND shielding!

AU\_VIN1\_N M3 AU\_VIN1\_N

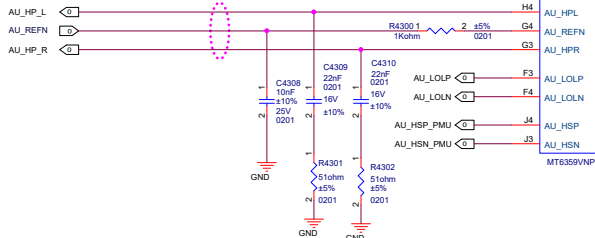
For P-N pair: differential pair & GND shielding!

AU\_VIN3\_P L4 AU\_VIN3\_P

AU\_VIN3\_N L5 AU\_VIN3\_N

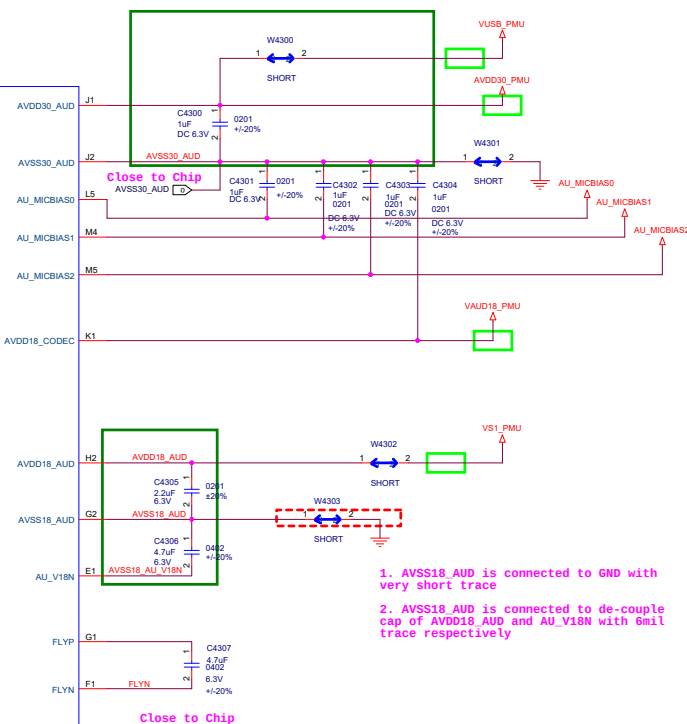
HP\_EINT K4 ACCDET

HP\_EINT K3 HP\_EINT



-AU\_HPL and AU\_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively

-The suggested layout pattern of AU\_HPL/ AU\_HPR/ AU\_REFN is " GND AU\_HPL AU\_REFN AU\_HPR GND"



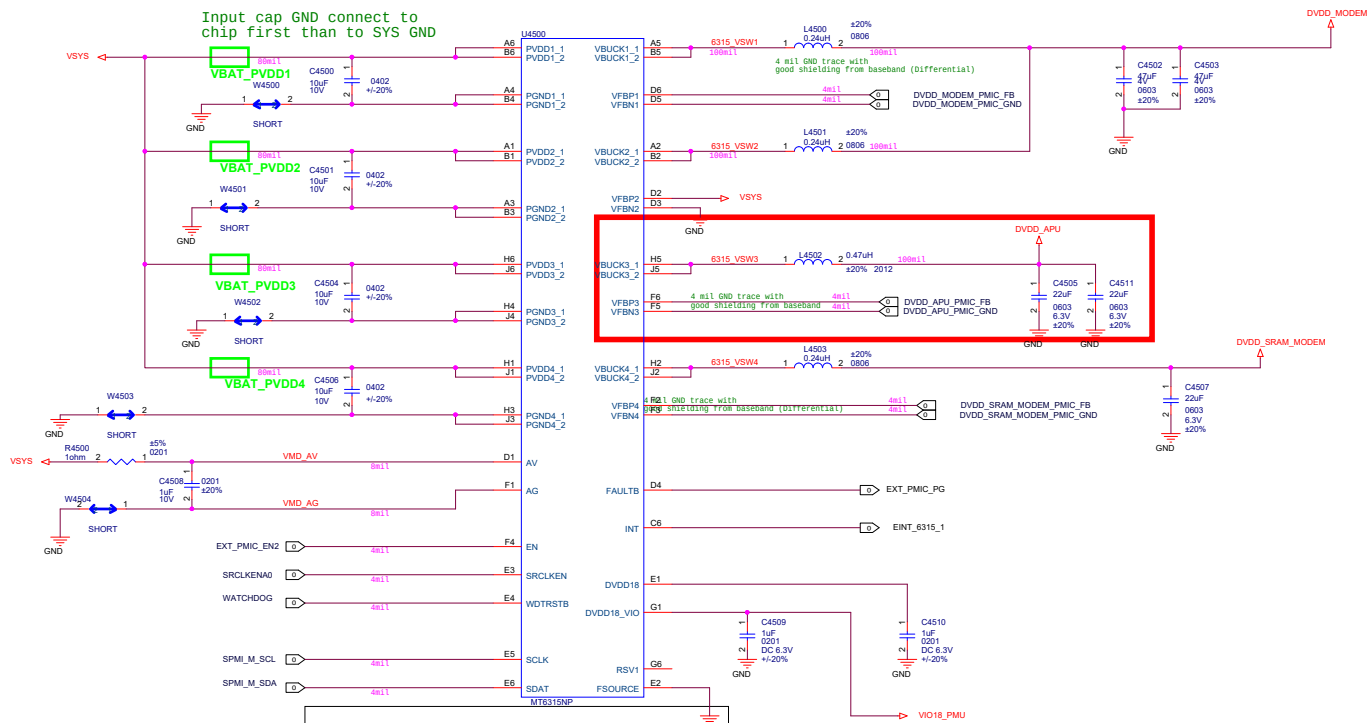
1. AVSS18 AUD is connected to GND with very short trace
2. AVSS18 AUD is connected to de-couple cap of AVDD18 AUD and AU\_V18N with 6mil trace respectively

Close to Chip

| MI PHONE |                         |                |
|----------|-------------------------|----------------|
| File     | MT6853                  |                |
| Size     | Document Number         | Rev            |
| A3       | MT6359_AUDIO            |                |
| Date:    | Monday, August 03, 2020 | Sheet 43 of 73 |



ALL C out C<sub>close</sub> to SOC

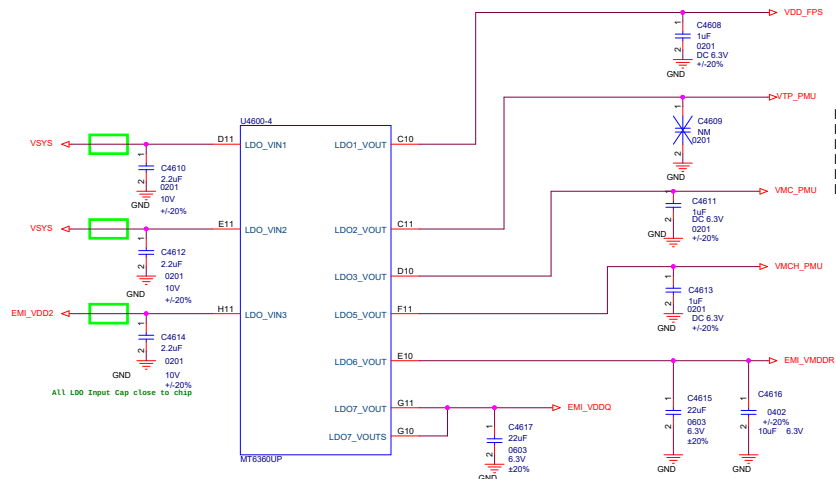
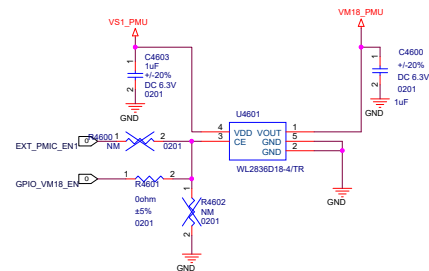
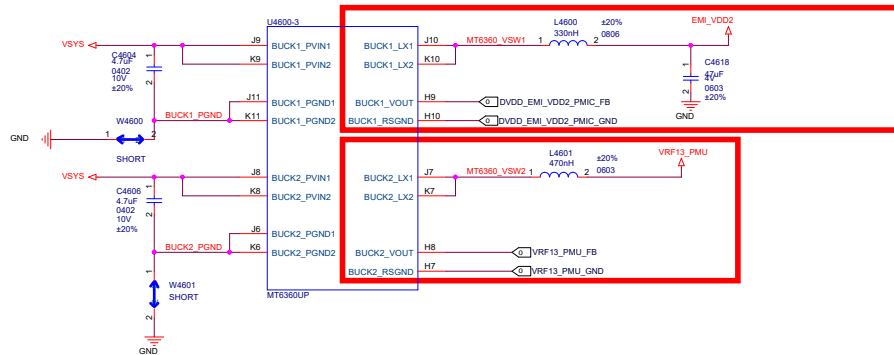


Note : Refer to Platform SoC Design Notice  
for PMIC part number

|              |                         |  |                |
|--------------|-------------------------|--|----------------|
| MI PHONE     |                         |  |                |
| Title MT6853 |                         |  |                |
| Size         | Document Number         |  | Rev            |
| A3           | MT6315_VMD              |  |                |
| Date:        | Monday, August 03, 2020 |  | Sheet 43 of 73 |

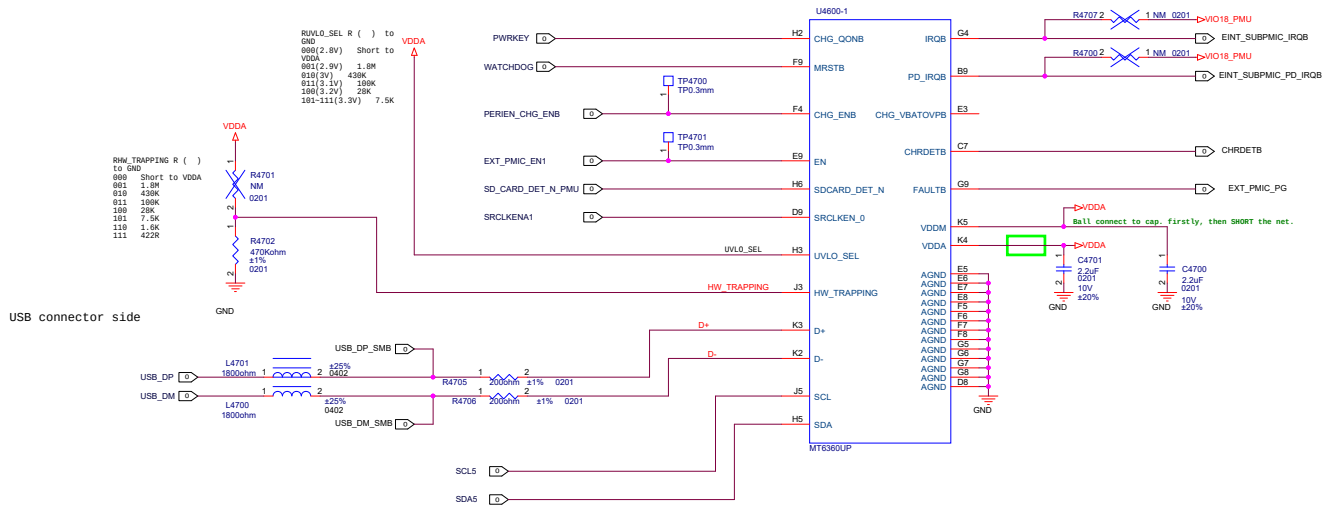


Note : Refer to Platform SoC Design Notice  
for PMIC part number

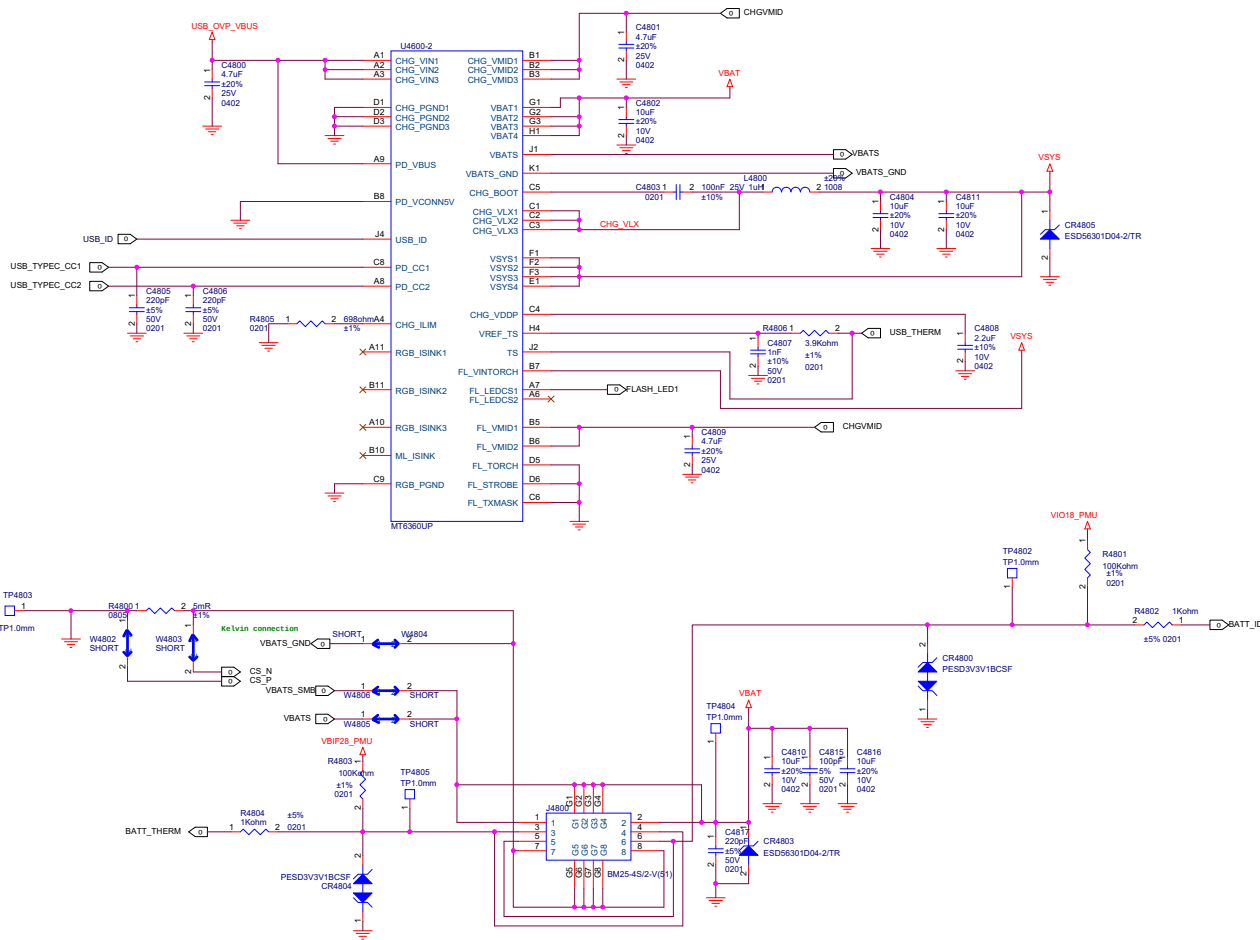


LD01 VFP  
LD02 VTP  
LD03 VMC  
LD05 VMCH  
LD06 VMDDR  
LD07 VMDDR\_EN

| MI PHONE |                         |       |          |
|----------|-------------------------|-------|----------|
| Title    |                         |       |          |
| Size     | Document Number         | Rev   |          |
| A3       | MT6360_BUCK_LDO         |       |          |
| Date:    | Monday, August 03, 2020 | Sheet | 46 of 73 |



| MT6853                  |  |  |  |
|-------------------------|--|--|--|
| Title                   |  |  |  |
| MT6853                  |  |  |  |
| Size                    |  |  |  |
| Document Number         |  |  |  |
| MT6853_General          |  |  |  |
| Date                    |  |  |  |
| Monday, August 03, 2020 |  |  |  |
| Sheet                   |  |  |  |
| 47 of 79                |  |  |  |



| MT6360UP |                         |                |
|----------|-------------------------|----------------|
| File     | MT6360UP                |                |
| Size     | Document Number         | Rev            |
| A3       | MT6360UP_Charger        |                |
| Date:    | Monday, August 03, 2020 | Sheet 48 of 79 |

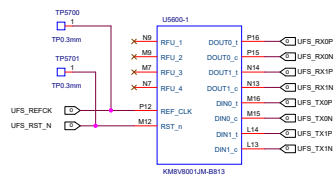




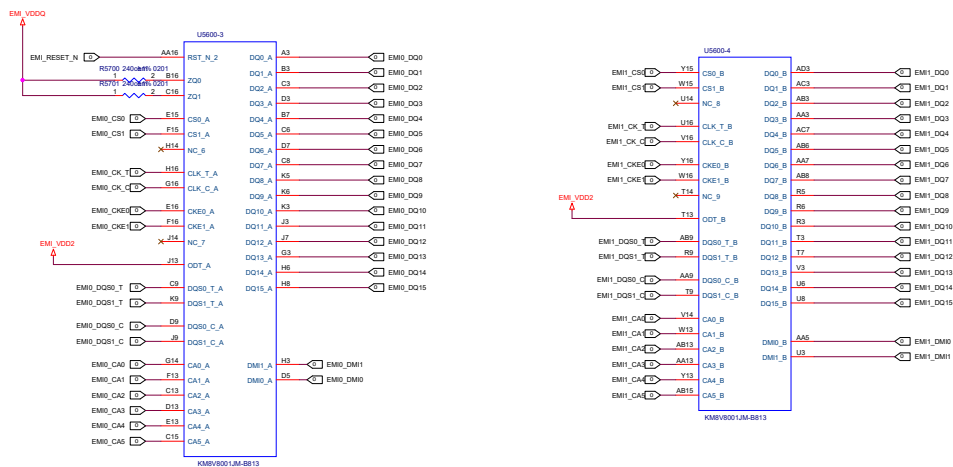




## UFS2.1



## LPDDR4/4x



|             |                         |  |                |
|-------------|-------------------------|--|----------------|
| MI PHONE    |                         |  |                |
| File MT6853 |                         |  |                |
| Size        | Document Number         |  | Rev            |
| C           | DDR4X                   |  |                |
| Date:       | Monday, August 01, 2011 |  | Sheet 67 of 73 |

Close to PMIC

Close to audio jack

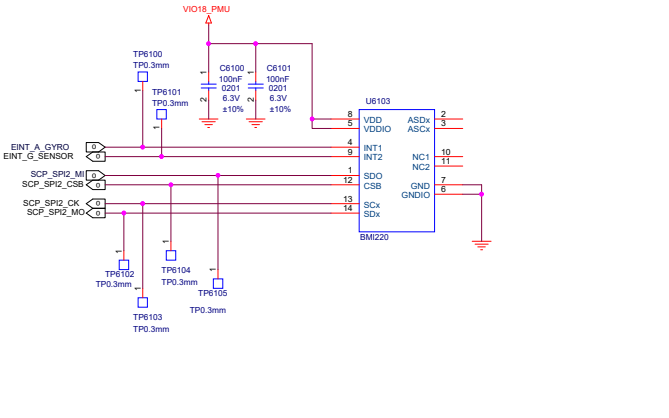
Connecting the GNDs together and then to main GND through signal via

# IC

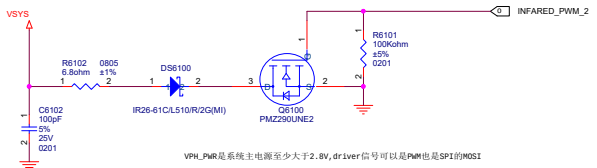
[illegible][illegible]

|                 |                          |        |    |           |
|-----------------|--------------------------|--------|----|-----------|
| MI PHONE        |                          |        |    |           |
| Title<br>MT6853 |                          |        |    |           |
| Size<br>A2      | Document Number<br>AUDIO |        |    | Rev<br>P1 |
| No.             | Quantity                 | Amount | On | Hand      |

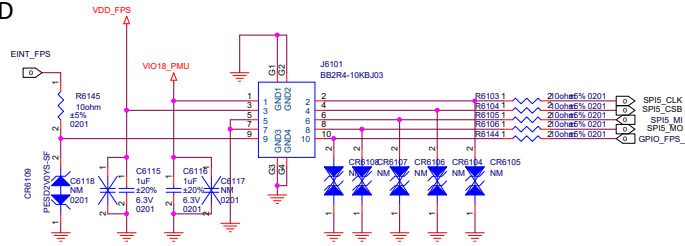
A+G



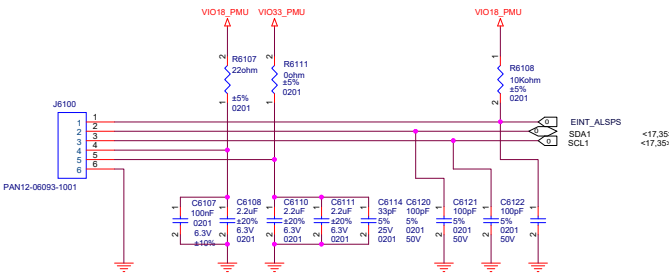
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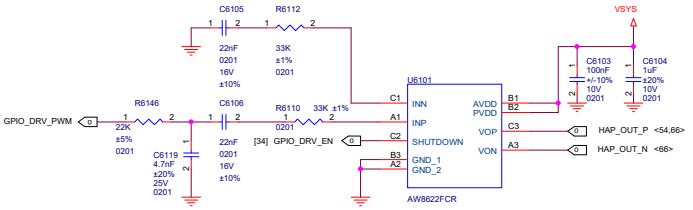
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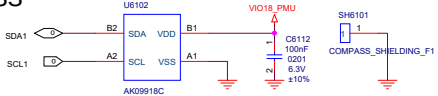
ALS+PS Connector



HAPTICS

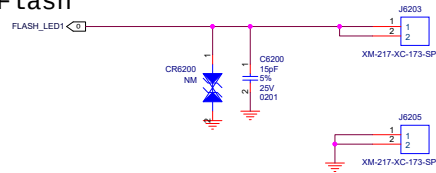


COMPASS

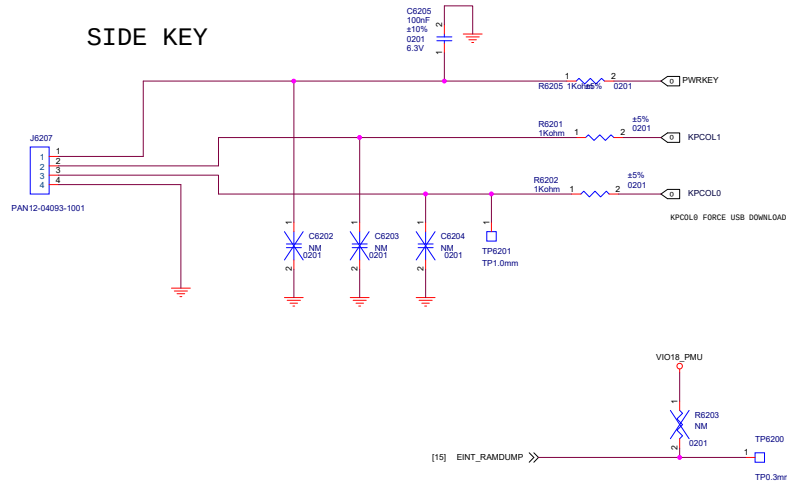


| MI PHONE                                    |  |  |  |
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| Title                                       |  |  |  |
| MT0853                                      |  |  |  |
| Size A3 Document Number SENSOR_1 Rev P0     |  |  |  |
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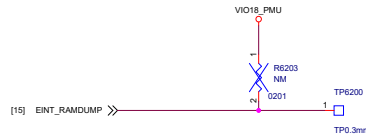
Flash



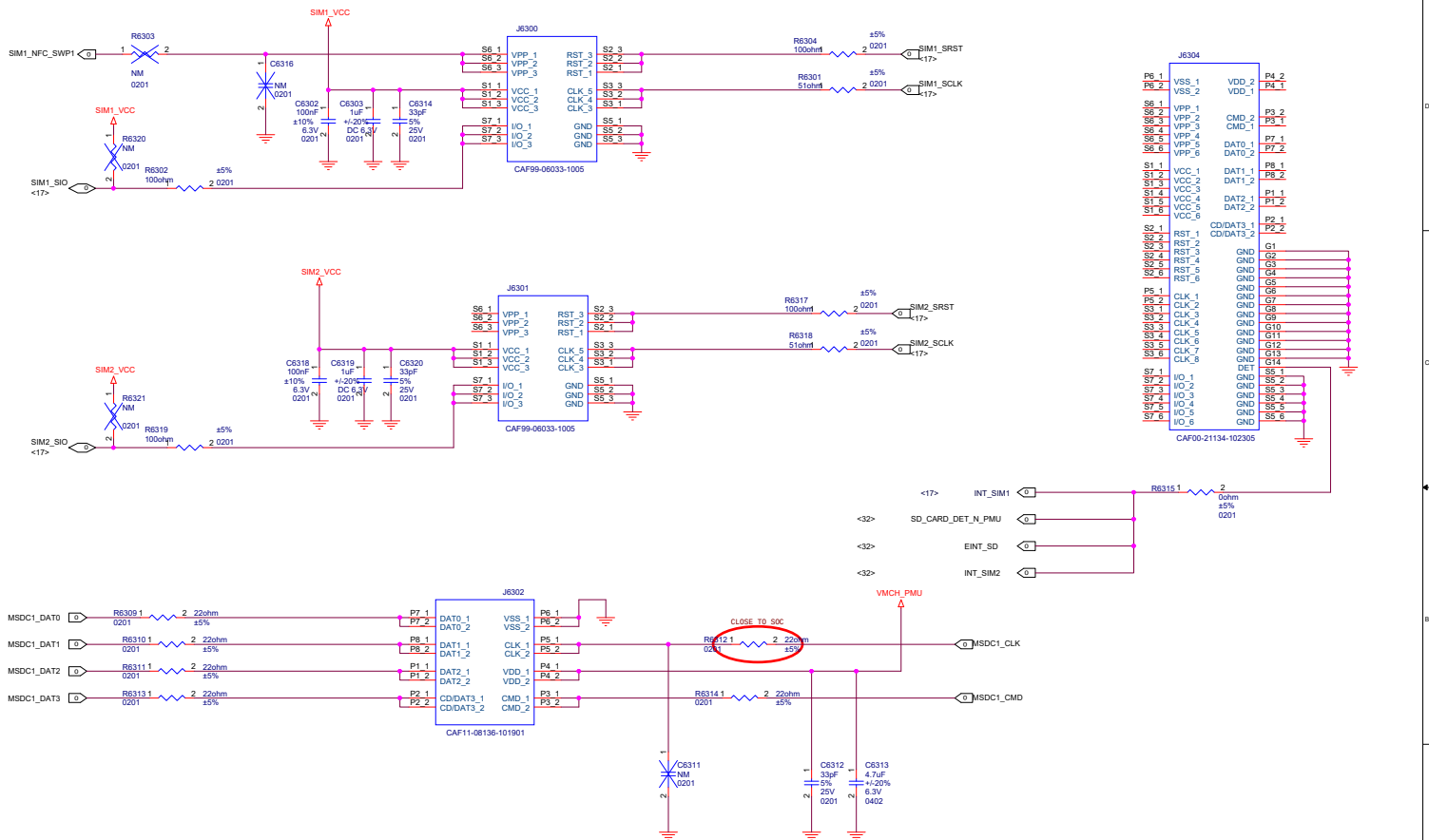
SIDE KEY



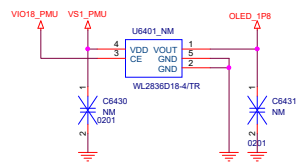
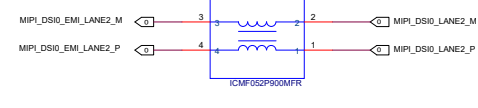
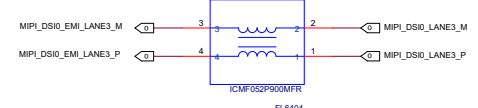
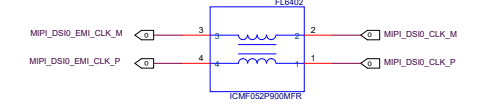
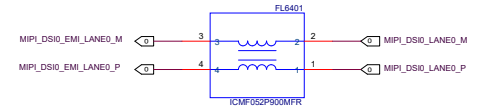
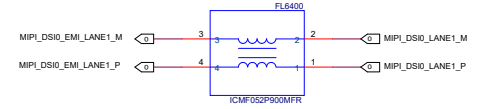
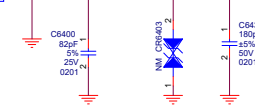
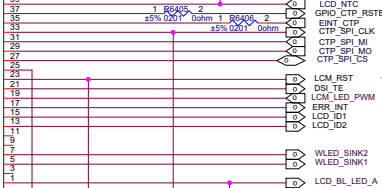
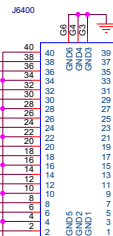
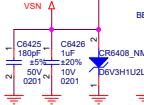
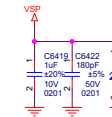
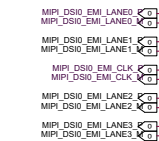
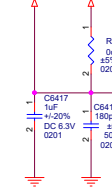
RAMDUMP debug key



| MI PHONE                |  |  |  |
|-------------------------|--|--|--|
| Title                   |  |  |  |
| MT6853                  |  |  |  |
| Size                    |  |  |  |
| A3                      |  |  |  |
| Document Number         |  |  |  |
| SENSOR-2                |  |  |  |
| Date:                   |  |  |  |
| Monday, August 05, 2020 |  |  |  |
| Sheet                   |  |  |  |
| 62 of 73                |  |  |  |
| Rev                     |  |  |  |
| PO                      |  |  |  |



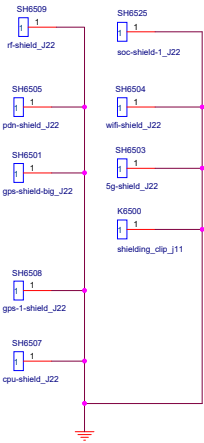
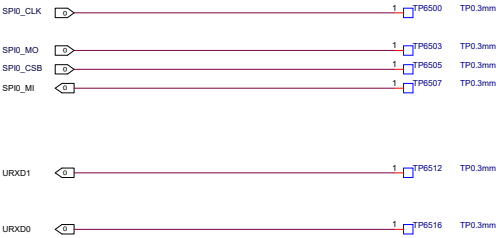
OLED\_1P8 VIO18\_PMU



| MI PHONE |                         |                 |          |
|----------|-------------------------|-----------------|----------|
| Title    | MT6853                  |                 |          |
| Size     | AS                      | Document Number | AMOLED   |
| Date:    | Monday, August 05, 2020 | Sheet           | 64 of 73 |

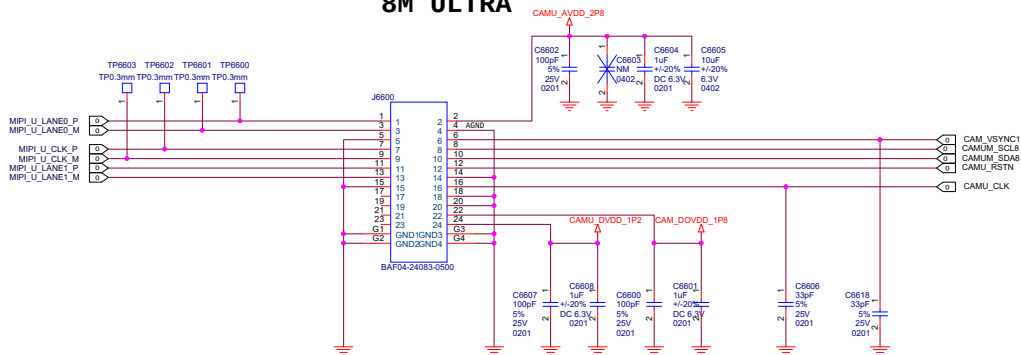
Schematic design notice of "90\_DEBUG\_IO" page.

Note 90-1: UART Default support : UART0,  
JTAG Default support : AP JTAG/MD\_IO\_JTAG

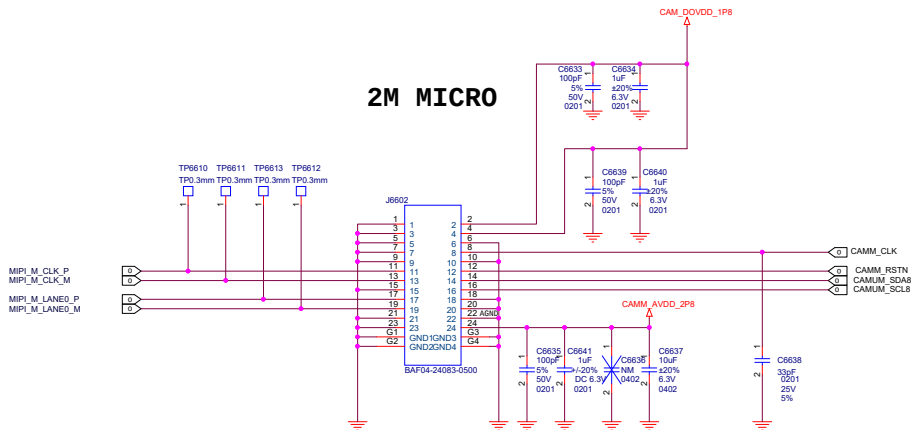


| MI PHONE |                 |  |                         |
|----------|-----------------|--|-------------------------|
| Title    |                 |  |                         |
| MT6853   |                 |  |                         |
| Size     | Document Number |  |                         |
|          | 90_DEBUG_IO     |  |                         |
| Date:    |                 |  | Monday, August 05, 2020 |
| Sheet    |                 |  | 65 of 73                |
| Rev      |                 |  | P0                      |

# 8M ULTRA

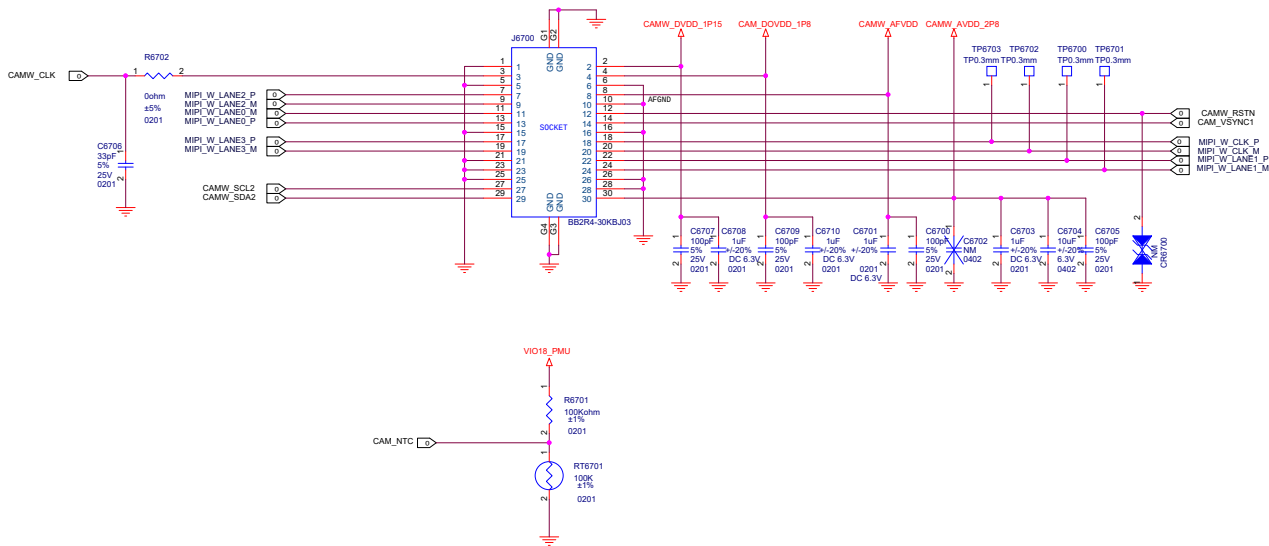


# 2M MICRO



| MI PHONE |                         |           |          |
|----------|-------------------------|-----------|----------|
| Title    | MT6853                  |           |          |
| Size     | Document Number         | Rev       |          |
| A3       | 8M MICRO & 8M ULTRA CAM | <RevCode> |          |
| Date:    | Monday, August 03, 2020 | Sheet     | 66 of 73 |



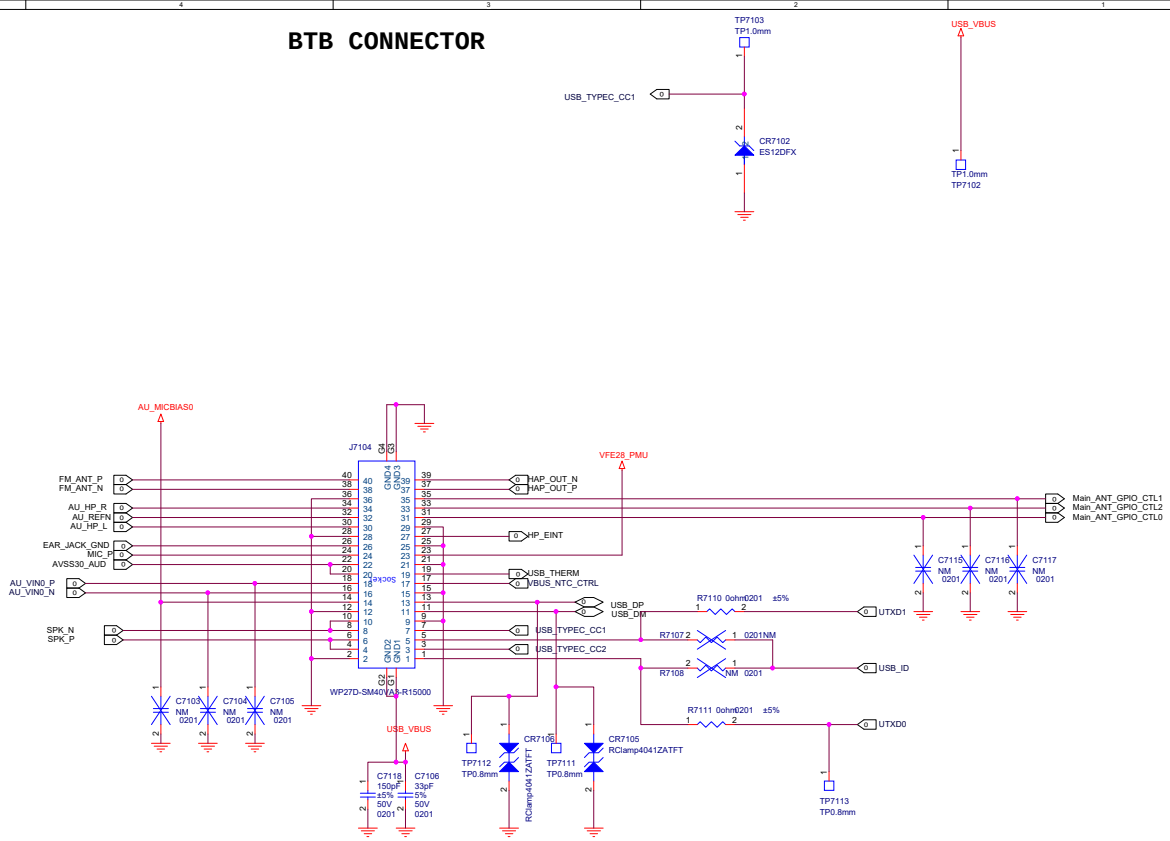


| MI PHONE |                         |       |          |
|----------|-------------------------|-------|----------|
| Title    | MT6853                  |       |          |
| Size     | Document Number         | Rev   |          |
| A3       | WIDE CAMERA             | P0    |          |
| Date:    | Monday, August 03, 2020 | Sheet | 67 of 73 |





## BTB CONNECTOR



|                |                                  |       |          |
|----------------|----------------------------------|-------|----------|
| MI PHONE       |                                  |       |          |
| File<br>MT6853 |                                  |       |          |
| Size<br>A3     | Document Number<br>USB CONNECTOR |       | Rev<br>A |
| Date:          | Monday, August 03, 2020          | Sheet | 71 of 73 |